MATH DOMINO STUDIO & 12-FRAME REGROUPING LAB

Triangular Combinatorics, Subitizing & Part-Whole Decompositions

STUDIO CONTINUUM: 1·Ten-Frames · **2·Dominoes** · 3·Clock · 4·Fractions · 5·Primes · 6·Algebra · 7·Probability · 8·Volvelle · 9·Quest · 10·Rack · 11·SC-32

CARD 02 / 10 • DOMI-01

MATHEMATICAL REFERENCE

1. WHAT IS THIS MANIPULATIVE?

TRIANGULAR COMBINATORICS & PART-WHOLE PAIRS

Double-six dominoes form a triangular math system.
It contains all unordered pairs [a|b] from 0 to 6:
Total Tiles = T(7) = (7 × 8) / 2 = 28 Unique Tiles
Each tile is a 2-part additive partition of 0–12!

MODULAR 28.0 × 14.0 MM TILE DATUM

- Tile Size: 28.00 × 14.00 × 7.00 mm (4U×2U×1U)
- Tactile Pips: 3.20 mm ⌀ recessed high-contrast cups
- Center Divider: Recessed 1.2mm scoring groove
- Pitch: 14.0 mm half matches Ten-Frame cell

12-FRAME REGROUPING TRAY & DOUBLES LID

Lid features doubles spine [0|0]–[6|6] and a regrouping tray to model part-whole sums!

DOMINO SUBITIZING & SUM: [5|4] + [3|2] = 14



[5|4] = 9 [3|2] = 5

PART-WHOLE ADDITION

[5|4] pips = 9
[3|2] pips = 5
Total: 9 + 5 = 14
T(7) = 7×8/2 = 28 Tiles
Double-Six Combinatorics

2. KIT PARTS INVENTORY (33 PIECES)

COMPONENT	QTY	SPEC / DIMENSION	PALETTE COLOR
Double-Six Tiles	28x	28.0 × 14.0 × 7.0 mm (T7)	Stealth Black
Wildcard Blanks	2x	28.0 × 14.0 × 7.0 mm [|]	Stealth Black
12-Frame Regroup Tray	1x	Dual 6-Bay Tactile Stage	Matte Charcoal
Doubles Organizer Lid	1x	7-Slot Doubles Spine Rail	Matte Charcoal
3x1 Storage Bin	1x	49.0 mm Height (7U)	Matte Charcoal

TOTAL: 30 Domino Tiles + 12-Frame Tray + Doubles Lid + Storage Bin = 33 Parts

HIGH-CONTRAST PIPS & TEXTURE

- Stealth Black Matte: Low glare, fingerprint resistant polymer.
- Brilliant White Pips: Standard dice quincunx configurations.
- Doubles Spine: Labeled slots [0|0] to [6|6] for organizing.
- High Durability: Scratch-resistant tactile surfaces.

3. WHY KEEP THIS REFERENCE CARD?

✓ NO LOST PIECE LEFT BEHIND

Classroom manipulatives get lost. Never abandon an incomplete set! Every individual math component is permanently stocked for single replacements.

Scan back QR or visit stackcalc.io/parts
Order individual replacement pieces anytime

✓ 10-SECOND SELF-CHECKING PACK-UP

Pack-up is a built-in math game! Arrange the 28 standard tiles and 2 wildcards into four neat columns of 7 and one row of 2 in the box floor. A missing tile breaks the smooth rectangular brick! Organizer lid snaps flush for silent zero-rattle carry.

✓ HANDS-ON TUTORIAL ON REVERSE

Flip this card over for a complete step-by-step hands-on lesson with concrete numbers, stage arrangements, and calculator cross-references!

Keep this card stored inside the box lid!

STACKCALC ARCHIVAL INSTRUMENT LABS

Designed in Mountain View, California, USA • Non-Toxic BPA-Free.
Classroom Lab Crates (10x / 30x Stations): stackcalc.io/schools

TRIANGULAR NUMBER COMBINATORIAL THEOREM

The number of unordered pairs chosen with replacement from n elements is T(n) = n(n+1)/2.

Universal 7.00mm Datum • Classroom Continuum (10 Kits)

OFFICIAL REFILL PORTAL

stackcalc.io/parts
Order replacement parts anytime

SERIES 1 • CARD 02 OF 10 • DOMAIN: DISCRETE COMBINATORICS & PARTITIONS • 7.00mm 1U PITCH • T(7)=28 TRIANGULAR TILES

LESSON GUIDE: TRIANGULAR COMBINATORICS & PART-WHOLE SUMS

Step-by-Step Hands-On Tutorial: T(7) = 28 Tiles & [5|4] + [3|2] = 14

Grades K–5 Addition & Combinatorics • Hands-On Classroom & Home STEM Studio Series

CARD 02 / 10 • DOMI-01

MATHEMATICAL GUIDE

SCAN FOR 3D VIDEO DEMO

OPEN CAMERA & SCAN

stackcalc.io/products/math-dominoes/

Zero app install • iPad & Chromebook

CURRICULAR ALIGNMENT

- CCSS.MATH.CONTENT.K.OA.A.3 (Decompose to 10)
- CCSS.MATH.CONTENT.1.OA.B.4 (Unknown Addend)
- TEKS 111.3.b.3.A (Model Addition)

Grades K–5 Addition & Combinatorics

MATHEMATICAL INQUIRY

If you flip a domino end-for-end, does the total pip sum change? Why or why not?

1 STEP 1: LINE UP DOUBLES SPINE [0|0] TO [6|6]

Place the 7 double tiles in the lid's organizer spine:
[0|0]=0, [1|1]=2, [2|2]=4, [3|3]=6, [4|4]=8, [5|5]=10, [6|6]=12.
These 7 doubles form the diagonal spine of the 28-tile triangle!

2 STEP 2: PAIR COMPLEMENTARY SUM-TO-6 TILES

Find tiles whose pip sum equals 6: [0|6], [1|5], [2|4], [3|3].
Place them in the 12-frame tray side by side.
Observe the inverse symmetry: as left side increases by 1,
the right side decreases by 1, preserving the constant sum 6!

3–4 STEPS 3 & 4: COMBINE TILES & REGROUP

3. COMBINE TWO TILES: Take tile [5|4] (sum 9) and [3|2] (sum 5). Seat in 12-frame tray.
4. CALCULATE SUM: 9 + 5 = 14! (1 full 10-frame + 4 ones).
Calculator Verification: Press Domino Pip Pattern Icon ([g][COUNT] & [g][a:b]) and run 5 [ENTER] 4 [+] 3 [ENTER] 2 [+] [+] → 14.

5 CORE PRINCIPLE: THE 28-TILE TRIANGULAR NUMBER SECRET

Why are there exactly 28 tiles? Count them by their highest pip: 7 tiles have a 0 ([0|0]..[0|6]), 6 new tiles have a 1 ([1|1]..[1|6]), 5 have a 2, down to 1 tile with a 6 ([6|6]).
Sum: 7 + 6 + 5 + 4 + 3 + 2 + 1 = 28! Dominoes are triangular numbers made tactile!

★ EXTENSION INVESTIGATION: FIND ALL SUM-TO-7 TILES

Can you find all 3 tiles that sum to 7: [1|6], [2|5], and [3|4]? Why is there no [0|7] tile in a double-six set? (Scan QR for 3D animated solution!)

→ Assemble with your physical blocks, then scan the QR code to verify with the 3D step-by-step vector model.

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Interactive Mathematics Manipulative Ecosystem.
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CROSS-REFERENCED CALCULATOR VERIFICATION

Every keystroke sequence verified with RPNCore CalculatorEngine.
Modular Hardware Continuum (9 Kits) • Universal 7.00mm Pitch • stackcalc.io

PARTS & SCHOOL REFILL PORTAL

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